

RAMKUMAR U

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LINKEDIN



GITHUB

SUMMARY

Full Stack Developer and IoT Software Engineer with experience in developing web, mobile, desktop, and IoT-based applications. Proficient in React, Node.js, Python (Flask), MySQL, and cloud platforms including AWS, Firebase, and Vercel. Experienced in industrial automation and IoT integration, including communication with PLCs, sensors, and IP cameras using TCP/IP and RTU-based protocols for real-time monitoring and control. Skilled in building scalable, secure, and user-centric solutions with strong problem-solving abilities and a passion for high-performance, modern application development.

EXPERIENCE

Innovative Web Solutions with Seilax Software Systems Pvt. Ltd.

Sep 2023 - Feb 2024

- Developed production-grade web applications including a Company Profile Website, Real Estate Portal, and Mall Leasing & Management System using React, Node.js, and Firebase.
- Built scalable React architecture using reusable components, centralized API services, and global state management, leveraging libraries such as MUI, Bootstrap, Framer Motion, Swiper, Axios, React Toastify, and XLSX.
- Developed secure RESTful APIs using Node.js and Express, leveraging Axios, Body Parser, CORS, dotenv, and Nodemon to ensure robust, scalable, and well-structured backend functionality.
- Live Projects: Company Website (www.seilaxsoftware.com) | Mall Leasing & Management Portal (portal.seilaxsoftware.com)

End-to-End Software & Industrial Automation Solutions – Bharath Vending Corporation

Feb 2024 - Present

At Bharath Vending Corporation, I worked as a Full Stack, IoT, and Automation Software Developer, responsible for designing and delivering end-to-end software solutions for industrial automation machines and smart vending systems. I developed web dashboards, desktop automation applications, and Android applications to control, monitor, and manage machines in real time. My work involved PLC-based automation, RTU (RS485/RS232) and TCP/IP communication, RFID and barcode-based workflows, IP camera integration, and printer handling, enabling secure, reliable, and scalable machine operations. I also handled backend architecture, database design, cloud deployment, and production server management on AWS EC2, including security groups, firewall configuration, reverse proxy setup, and domain handling. These solutions were built to ensure automation accuracy, operational transparency, anti-theft control, and system scalability across multiple industrial use cases.

PROJECTS

1. Caterpillar Decal Vending Machine Software

- Built an end-to-end IoT vending solution with a web dashboard and desktop automation app, enabling RFID & barcode-based automatic decal dispensing, role-based access, and real-time machine monitoring.

2. Milk Weight Monitoring & Anti-Theft System

- Developed a dashboard and Android application to track vehicle and BMC-wise milk weights in real time, generate receipts via Bluetooth printers, and prevent theft through remote weight validation.

3. Advanced Vertical Storage System (ASRS)

- Implemented an automated storage and retrieval system with role-based access, inventory tracking, low-stock email alerts, and desktop-controlled industrial automation.

4. Orange Juice Automation Machine

- Built a fully automated juice vending system with desktop machine control, stock management, and PhonePe payment integration for cashless ordering and dispensing.

5. Part Validation & Quality Inspection Software

- Developed an industrial part validation system using PLC integration and IP cameras, enabling QR-based part tracking and automated visual inspection for quality assurance.

EDUCATION

BE in Mechatronics Engineering

M.A.M School Of Engineering . Trichy, India (2023)

SKILLS

- **Full Stack Web Development:** Strong experience building scalable web applications using React.js and Node.js, leveraging frontend libraries such as MUI, Bootstrap 5.3, Framer Motion, Swiper.js, Axios, React Toastify, React TS Particles, React Player, XLSX, and AOS, along with backend technologies including Express.js, Axios, Body Parser, CORS, dotenv, Nodemon, Nodemailer, and template engines such as EJS, Pug, and Handlebars, implementing centralized API services, reusable components, and global state management.
- **Desktop & Mobile Application Development:** Developed cross-platform desktop applications (EXE) using Electron (electron-builder, auto-launch, updater, log, is-dev, fs-extra, node-fetch, node-wifi, dotenv) and Android applications (APK) using Capacitor (@capacitor/core, cli, android, assets), including auto-updates, system controls, and automation workflows.
- **Industrial Automation & IoT Integration:** Hands-on experience with TCP/IP and RTU (RS485/RS232) communication, integrating PLCs, IP cameras, RFID readers, barcode scanners, receipt and barcode printers, using Python (Flask) as automation middleware with libraries such as pymodbus, OpenCV, Flask-JWT-Extended, SQLAlchemy, python-barcode, Pillow, pywin32, pandas, numpy.
- **Database, Cloud & Deployment:** Proficient in MySQL database design and data handling, and cloud deployment on AWS and Firebase, with deep experience in AWS EC2 including server provisioning, security groups, firewall and network configuration, reverse proxy and domain setup, and frontend deployment using AWS Amplify, along with Firebase hosting and deployment. Experienced in Windows IIS deployment, including application hosting, site configuration, bindings, SSL setup, and environment configuration for production environments.